# A LOCALLY FINITE TYPE MORPHISM OVER A NAGATA BASE HAS NAGATA SOURCE

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The author's proof contains the slip of writing V for U in the sentence 'Hence V is a Nagata scheme by Morphisms, Lemma ...'; the text is delivered verbatim rather than corrected, since no rewriting of author text is permitted.

*Editorial note.* The single reference of the counted spans is recorded as a non-load-bearing navigation pointer; the argument that consumes it stays inside the counted proof span.

*Editorial note.* Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

*Allowed standard prerequisites (editorial).*

- Nagata-ness of an algebraic space is etale local. used at proof lines 5245-5248. Makes V a Nagata scheme from the hypothesis on Y.
- Local finite typeness is read on a presentation. used at proof lines 5249-5250. Makes U -> V locally of finite type.
- The scheme case: locally finite type over a Nagata base has Nagata source. Morphisms, Lemma morphisms-lemma-finite-type-nagata, used at proof lines 5250-5251. Gives the Nagata property for the cover of X.
- Nagata-ness of X read off from a presentation. used at proof lines 5252-5252. Concludes for X.

*Source reference.* Original lemma 035A: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4121–4121. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

**Lemma 0.2.** *Let $S$ be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over $S$. If $Y$ is Nagata and $f$ locally of finite type then $X$ is Nagata.*

*Proof.* Let $V$ be a scheme and let $V \to Y$ be a surjective étale morphism. Let $U$ be a scheme and let $U \to X \times_Y V$ be a surjective étale morphism. If $Y$ is Nagata, then $V$ is a Nagata scheme. If $X \to Y$ is locally of finite type, then $U \to V$ is locally of finite type. Hence $V$ is a Nagata scheme by Morphisms, Lemma 035A. Then $X$ is Nagata by definition. □

*Date*: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.